



Review

Understanding de-inking in packaging plastic recycling: Bridging the gap in resource conservation and establishing average hazard quotient

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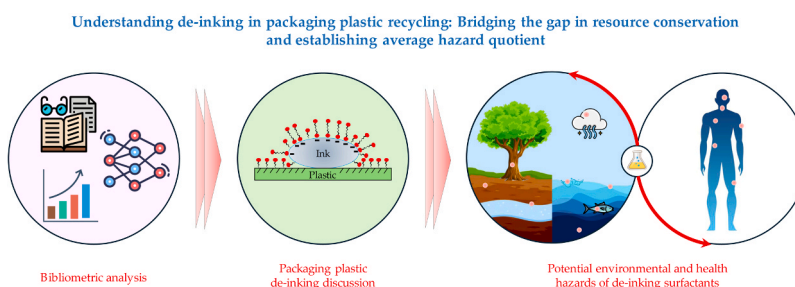
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HIGHLIGHTS

- De-inking is an undermined pre-treatment step in packaging plastic recycling.
- There are concerns over the toxicity and hazards of surfactants used in de-inking.
- This study critically analyses the state of the art of packaging plastic de-inking.
- Developed a chemical selection tool that minimises hazards in three categories.
- Bio-based and green chemicals should be explored for packaging plastic de-inking.

GRAPHICAL ABSTRACT



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ABSTRACT

Achieving circularity in the plastic economy predominantly depends on sourcing higher quality recyclates. Packaging plastic poses a significant challenge as it is often not prioritised for collection or recycling initiatives. The presence of additives, such as printing ink, impedes the quality of recyclates. Considering the volume of packaging plastics and the importance of branding (aesthetics and consumer information), ink removal is a critical pre-treatment step. However, the literature is limited, with only 14 studies exploring de-inking processes. Drawing parallels with the detergent laundering process, surfactants have been widely investigated in plastic de-inking, with cationic surfactants proving the most effective with a de-inking efficiency of up to 100%. However, concerns exist regarding the toxic and hazardous nature of the surfactants and chemicals. The average hazard quotient (AHQ) was developed, which compares de-inking chemicals as one of the key findings. AHQ provides a quantitative proxy for the hazards and toxicities, which are qualitatively presented as part of the globally

Abbreviations: AHQ, Average Hazard Quotient; CMC, Critical Micelle Concentration; GHS, Globally Harmonised System; GVL, Gamma-Valerolactone; HDPE, High-Density Polyethylene; HLB, Hydrophilic-Lipophilic Balance; LAS, Linear Alkylbenzene Sulfonates; LCA, Life Cycle Assessment; LDPE, Low-Density Polyethylene; LHV, Lower Heating Values; MSDS, Material Safety Data Sheets; NC, Nitrocellulose; OPV, Over-Print Varnishes; OpenCV, Open-Source Computer Vision; PE, Polyethylene; PET, Polyethylene Terephthalate; PP, Polypropylene; PU, Polyurethane; PVB, Polyvinyl Butyral; PZC, Point of Zero Charge; SDS, Safety Data Sheets; STRAP, Solvent-Targeted Recovery and Precipitation; UV, Ultraviolet; WoSCC, Web of Science Core Collection.

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harmonised system (GHS) classification of chemicals. To drive emerging packaging plastic de-inking, including the development of green surfactants (e.g. gamma-valerolactone), this work enables an informed chemical selection minimising potential hazards (rather than creating more adverse effects in plastic recycling processes) and toxicities from plastic waste, fulfilling the objectives of cleaner plastic waste recycling.

1. Introduction

Since the early 1900s, the annual plastic production has grown 230-fold to reach 460 million tonnes in 2019 [1]. Two-thirds of the plastic waste generated in 2060 will be from ephemeral applications such as packaging, consumer products and textiles [2]. Packaging plastic accounts for 44% of polymer consumption [3] and 46% of plastic waste generation [4]. By 2060, the sector is projected to double in size, more specifically, the use of polyethylene terephthalate (PET) and polyethylene (PE) [1]. The growth is primarily from the versatile use of PE as short-lived packaging plastic in flexible packaging applications [5]. Even with a high recycling potential, an effective recycling strategy for flexible packaging waste is lacking. This can be attributed to material properties and additives, lack of technology readiness, and inadequate material recovery strategies [6,7]. Due to this, 95% of the packaging material, valued between USD 80–120 billion, is lost from the economy annually, and the resultant additional recoupment cost to society per annum is USD 40 billion [8].

Packaging plastics are often printed with inks as part of branding or to convey necessary consumer information. On average, 2–4% of the total weight of packaging plastic is printed ink [9]. While these inks are essential, they are also the source of significant contamination in plastic recycling [10,11]. Printing inks are mainly composed of pigments that impart colour, binders that provide adhesion, carriers (also referred to as solvents) that hold the suspension and additives for additional characteristics (Table 1).

Based on the carrier, inks are classified as water-based, solvent-based or UV-based (Table 1) [12]. The carriers evaporate during ink drying processes (i.e. manufacturing stage) and are insignificant during recycling [13]. However different binders are used, based on the carrier type, which is problematic in packaging plastic recycling. Solvent-based inks are compatible with most surfaces on which to print. In contrast, water-based inks have limited application as water evaporates slower than solvent, making the drying process energy-intensive. These inks are preferably used in papers and boards, which promote absorption capabilities. Finally, UV-based inks use reactive resins such as acrylates that can be activated by UV irradiation [14,15].

Printing inks leads to several issues when recycling packaging plastic waste [9], and mixing plastics with different colour pigments during recycling results in black, brown or grey recyclates [16]. These are often suitable only for downcycled applications and have lower market value [17,18]. The presence of inks was also found to lower the tensile properties of recyclates [13]. Further, nitrocellulose (NC), one of the most widely used binders in flexible packaging plastic (Table 1), is vulnerable to thermal degradation between 160 – 185 °C. Since extrusion temperatures for mechanical recycling could go up to 300 °C, NC

degrades into CO₂ and NO₂ at these temperatures, causing ‘gassing’ and the brownish colour. While other ink binders do not follow similar gassing patterns, they are also susceptible to decomposition at extrusion temperatures [9,19]. This shows the importance and benefits of de-inking packaging plastic waste before conventional recycling.

Realising a circular plastics economy will significantly depend on advancements in the recycling infrastructure, especially closed-loop recycling strategies. In such cases, removing additives and contaminants becomes imperative to producing higher-quality recyclates [20]. De-inking is a critical yet undermined pre-treatment step in recycling printed plastic waste. In the past decade, a renewed focus has been on de-inking research. Studies have explored the potential of various de-inking media, specifically surfactants. The effect of surfactant concentration, pH, temperature, ink pigment composition, and other parameters in de-inking have also been experimented with. However, there is a lack of streamlined information on how the research area has developed over time and the status quo of plastic de-inking. Thus, a systematic literature survey will be conducted to collate and critically analyse the packaging plastic de-inking literature. In addition, the surfactants and chemicals employed in previous de-inking studies have not been assessed for their toxic and hazardous nature. Thus, as another niche aspect, a chemical selection tool based on the average hazard quotient (AHQ) will be formulated, and it will offer new avenues for sustainable de-inking surfactant and chemical development in the future.

2. Background info on plastic waste recycling

Plastic waste can be managed through closed-loop or open-loop recycling (Fig. 1). In closed-loop recycling, plastic waste is recycled back into plastic products in the same product category, though open-loop recycling produces a recyclate employed in different applications.

2.1. Mechanical recycling

Mechanical recycling (cascade recycling) is the most prevalent method employed at a large scale. This is further classified into primary and secondary recycling based on the source of waste [21]. Mechanical recycling is affected by the quality of the waste stream, as different types of plastic respond differently to treatment based on their physio-thermo-chemical properties [22,23].

Primary recycling sources plastic waste from post-industrial applications, which are of higher quality. The collected waste is often recycled back into production after shredding and blending with virgin plastic. Secondary recycling handles the post-consumer plastic waste mixed with different plastic types (i.e. lower quality feed) [24]. The key

Table 1
Typical composition of printing inks on packaging plastic.

Component	Common chemical composition		Typical content (wt%)
Carriers	Water-based	Water with alcohol	50 – 70
	Solvent-based	Organic solvents (e.g. alcohols, esters, ketones)	
	Ultraviolet (UV)-based	Liquid with UV reactive substances	
Binders	Water-based	Methyl methacrylate, styrene, butyl acrylates, acrylic and methacrylic acids	20 – 30
	Solvent-based	Nitrocellulose (NC), polyamide, polyvinyl butyral, polyvinyl chloride, polyurethane	
Pigments	Organic and inorganic substances (e.g. PbK7)		6 – 30
Additives	Wax, surfactants, dispersants, and crosslinkers		1 – 10

Source: Adapted from [12,13].

steps involve washing, shredding, melting and remoulding. In secondary recycling, the quality of the recyclates is inferior, resulting in down-graded applications. Other interferences, such as organic contaminants, labels, inks, fillers, and additives, significantly deteriorate the material and optical properties of the recycled product [25,26]. Thus, secondary recycling is not favourable for a circular plastic economy in the long run.

2.2. Waste-to-energy recycling

In this open-loop recycling process, the heat from incinerating plastic waste generates superheated steam for electricity production [27]. The lower heating values (LHVs) of PE (42.8 MJ kg^{-1}) and polypropylene (PP) (46.5 MJ kg^{-1}) are closely comparable to that of petroleum (42.3 MJ kg^{-1}) and kerosene (46.5 MJ kg^{-1}) [28]. However, post-consumer plastic waste (mixed, low-quality) is 40% less efficient, which can be improved upon segregation. It has been shown that the best fuel efficiency is obtained from the purest waste streams [27]. A comparative study by the Waste & Resources Action Programme (WRAP) showed that recycling was better than incineration, considering greenhouse gas emissions and energy balances [29]. In contrast, incineration performed better when a significant volume of recycled plastics could not be incorporated back into virgin plastic production or when there was a high washing requirement [29]. Even if waste-to-energy is prioritised higher than landfilling in the waste hierarchy, its preference is debated. This is because waste-to-energy provides an incentive to avoid other recycling strategies where the materials could be retained in use for longer life. Hence, it contradicts the principles of the circular economy.

2.3. Chemical recycling

The chemical recycling method comprises various strategies to convert plastic waste into reusable monomers or to produce fuels and feedstock for subsequent processes [21]. 'Chemical recycling' and 'feedstock recycling' are often used interchangeably. In reality, chemical recycling can be considered an umbrella term including both open-loop thermolysis and closed-loop solvolysis. Thermolysis, also known as 'feedstock recycling', produces chemicals for other purposes, including fuels, syngas or naphtha. Whereas solvolysis produces monomers that can be used in polymer production [30]. Chemical recycling is limited by economic viability, feedstock availability and feedstock quality. Often, it is hard for the recycling process (including pre-treatment and purification steps) to compete with the costs of virgin material production. Heterogenous plastic waste from post-consumer sources, contaminated with additives, colourants, impurities, moisture and dirt, severely impacts the feedstock quality. Nevertheless, extensive research is focused on chemical recycling due to the potential to recover high-quality recyclates and remove contaminants, such as halogenated flame retardants [27]. It was also found that supplementing mechanical and chemical recycling can improve the overall recycling rates [27]. Holistically, advancements in chemical recycling are needed to maximise material utilisation and further circularity in the plastic economy.

3. Methodology

To comprehensively understand the field, literature on packaging plastic de-inking was collected following the steps detailed in Sections 3.1–3.3. A bibliometric study is conducted to analyse the evolution of the research field. This is followed by the methodology for the chemical selection tool of de-inking surfactants and chemicals in terms of the AHQ (Section 3.4).

3.1. Search strategy

The relevant publications in the field are gathered from the three major scientific databases – Scopus, Web of Science Core Collection (WoSCC) and Google Scholar. The following search strings were used in each case:

Scopus: TITLE-ABS-KEY (de-inking OR deinking OR "ink removal") AND TITLE-ABS-KEY (plastic OR "plastic waste" OR "packaging waste").

WoSCC: (TS=(de-inking) OR TS=(deinking) OR TS=("ink removal")) AND (TS=(plastic) OR TS=("plastic waste") OR TS=("packaging waste")).

Google Scholar: (de-inking of plastic waste).

Boolean operators (OR, AND) were used in the search string to include common usages and emphasise the relation between the key topics, such as de-inking and plastic waste. All results available until 3rd September 2023 were collected.

3.2. Study selection

The relevant literature to be included in the study was identified through a four-step screening process, as explained below:

Step 1 – The language of the publications was restricted to English.

Step 2 – The type of literature was limited to peer-reviewed journal publications.

Step 3 – The title and abstract of each publication were screened to identify relevant literature.

Step 4 – Duplicate publications were removed.

However, a slightly different screening process was followed for Google Scholar. This is because an exhaustive list of publications from paper de-inking overlapped with the current search results.

Step 2 – Publications related to paper de-inking were removed by adding the search string (-wastepaper -newspaper -"deinking sludge" -"paper deinking").

Step 3 – The type of literature was limited to peer-reviewed journal publications. The title and abstract of each publication were screened to identify relevant literature.

In addition, the reference lists of all the selected publications from the database search were scoped to find any other relevant publications to be included. The screening process is presented in Fig. 2.

3.3. Data extraction and analysis

For the bibliometric analysis, an open-source science mapping tool

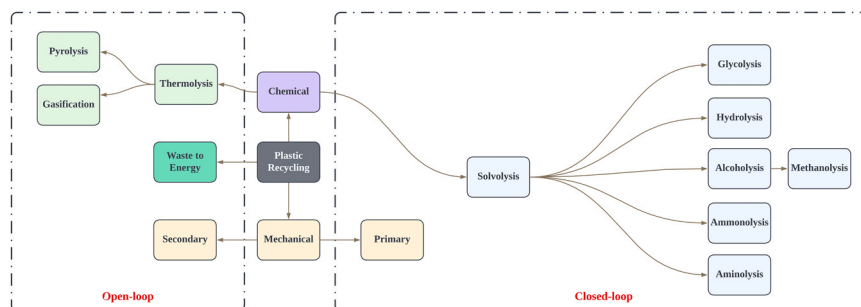


Fig. 1. The available process routes for recycling plastic waste.

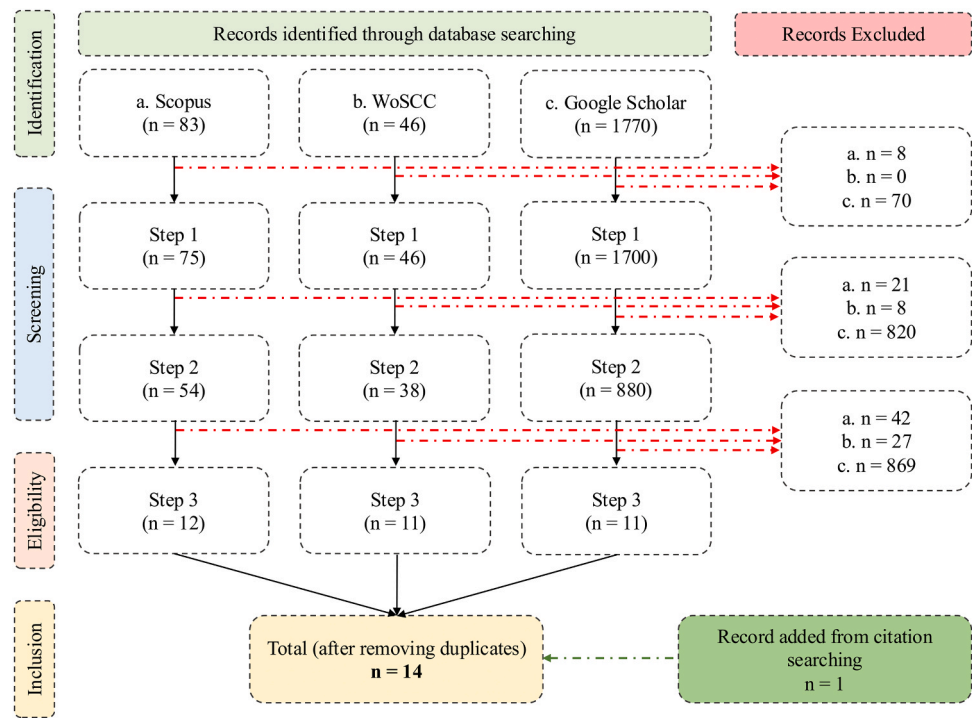


Fig. 2. Flowchart for the literature screening process for packaging plastic de-inking.



Fig. 3. The summary of bibliometric analysis on the collected packaging plastic de-inking studies.

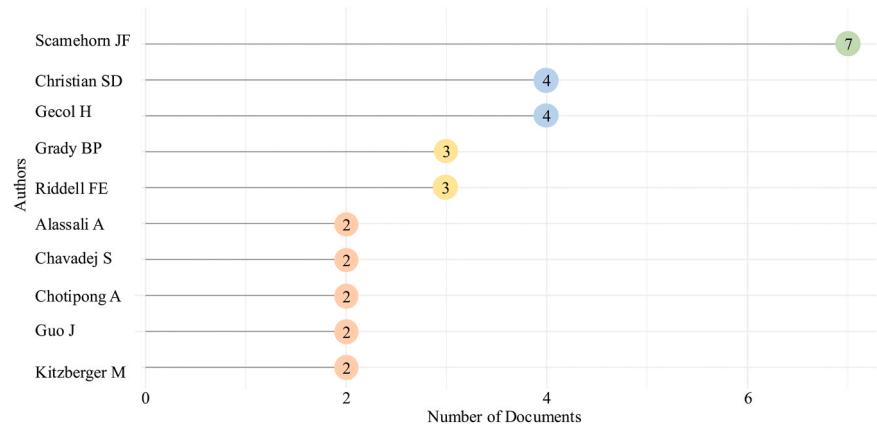


Fig. 4. Chart showing the relevance of authors who performed packaging plastic de-inking studies.

called Bibliometrix [31], which utilises the biblioshiny package in RStudio, was used. A summary of the study is provided in Fig. 3. A total of 14 publications were identified based on the inclusion criteria. A detailed summary of the 14 studies is provided in Table D.1 based on the plastic and ink types studied, characterisation and de-inking tests conducted, the various de-inking parameters and the resultant de-inking efficiencies. The field is relatively underexplored, with the first research published in 2001. Over the years, 70 researchers have contributed to the area, with an annual growth rate of 6.5%.

The field was pioneered with the works of John F Scamehorn's collaborative research team. Between 2001 and 2007, he authored seven papers exploring cationic, anionic, and non-ionic surfactants for removing water- and solvent-based inks. Therefore, he is identified as the most relevant author in de-inking literature to date (Fig. 4). The top 4 most relevant authors in the field after Scamehorn J.F. are Christian S. D., Gecol H, Grady B.P., and Riddell F.E. It is to be noted that they are all from the same research collaboration, sharing co-authorship for most of the papers. The most relevant affiliations in the field are the University of Oklahoma, the University of Nevada and Chulalongkorn University.

This explains why Gecol et al. [32], the first paper on de-inking, is the most cited document globally (36 citations as of 18th January 2024). This study explored the potential of various surfactants in de-inking water-based inks from PE films. The de-inking efficiency is studied by varying pH and critical micelle concentrations (CMC). Additional bibliometric discussion is provided in Appendix A.

3.4. Development of AHQ model

The surfactants currently used in de-inking are toxic and hazardous. However, there may be instances where such surfactants are unavoidable. The previous de-inking studies did not quantify the surfactants' physical, environmental and health hazards. Furthermore, no specific toxicity tests were performed using standards, such as the Globally Harmonised System (GHS) and the Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) of the European Parliament and of the Council of 18 December 2006 (EC No 1907/2006) [33].

Traditionally, the methods of hazard classification follow a qualitative or semi-quantitative approach, which provides the necessary information but can be complex to interpret and compare chemicals directly. This presented the need for a mathematical model to categorise and rank the chemicals employed in de-inking based on their potential hazards. As such, the AHQ was introduced to simplify comparisons and decision-making on surfactant and chemical selection.

The GHS hazard classification provides baseline data to model the AHQ in this work. As of June 2023, twenty-one countries have fully adopted the GHS classifications, including Australia and Malaysia [34]. Several countries have adopted part of it as required, as well. The GHS classifications are presented as part of material safety data sheets (MSDS) or safety data sheets (SDS) for chemicals. They offer a standardised approach to presenting hazard classification and necessary precautions. However, GHS only provides qualitative data on hazards by broadly classifying them into three classes: physical, health, and environmental. There are 16 physical hazards, 10 health hazards and 2 environmental hazard classes. Based on the severity, each hazard is categorised from 1 to N (where N = 2, 3, 4, etc.) or A to G. In each case, 1 or A presents a more severe hazard, and severity reduces as the value increases. Furthermore, every SDS has a signal word, such as "danger", "warning", or "nothing". It was observed that the signal word "danger" was used whenever a category 1 or 2 hazard was present in any of the classes.

This work employed the given classifications and qualitative descriptions to model the AHQ (Eq. 1). A sample AHQ calculation using SDS data for cyclohexane is shown in Appendix B.

$$AHQ = \sum_{\alpha} r_{\alpha} \left[\frac{1}{N_{\alpha}} \times \sum_{i=1}^n \frac{1}{H_i^f} \right] \quad (1)$$

where α is the hazard class (physical, health or environment), N is the number of hazards in each class as shown in the SDS, H is the category of hazard (1, 2, 3, A, B etc.), f is the significance factor for categories (if H is 1 or 2, then $f = 1$; if $H \geq 3$, then $f = 2$), and r is the significance factor for hazard classes (by default, $r = 1$ to assume equal priority for all hazard classes unless specified otherwise, and $r_{\text{physical}} + r_{\text{health}} + r_{\text{environmental}} = 3$).

4. Packaging plastic de-inking

4.1. De-inking methodology

Based on the literature, a generalised de-inking procedure can be summarised as follows:

1. Sample preparation: Commercial PE films [32,35], uni-coloured PE films [36], PP films [11], rigid high-density polyethylene (HDPE) [37,38], low-density polyethylene (LDPE) films [13], and multilayer plastic films [39] were used in previous studies. The samples are not typically derived from post-consumer plastic waste, except in Guo et al. [12] and Sánchez-Rivera et al. [39], who used post-consumer and post-industrial waste, respectively. The printed packaging plastic is cut into small pieces as feasible for the test runs. In addition, some pre-treatment operations, such as pre-soaking and abrasion, were also employed.
2. Surfactant preparation: The required amount of surfactant is determined based on the surfactant-specific CMC and the test condition (above, below or at CMC). The surfactant solution is freshly prepared by mixing the required surfactant with either de-ionised or distilled water. The pH level of the surfactant solution is immediately adjusted using either sodium hydroxide or hydrochloric acid solutions.
3. Experiment: The cut plastic pieces are soaked in water for the specified duration to analyse the effects of pre-soaking. The samples are then transferred to the surfactant solution in a beaker in a temperature-controlled water bath or magnetic stirrer. The experiment is allowed to run for four hours on average. Once the experiment is completed the samples are thoroughly washed and air-dried. Some researchers have employed control experiments in de-inking, for example, using water [13,32]. However, those did not result in any significant de-inking.
4. Image analysis and de-inking efficiency: Using CIELAB, flake sorting or scanning and pixel reading methods detailed in Appendix C, the samples are analysed before and after the experiment. The respective equations are then utilised to calculate the de-inking efficiency in each case.

4.2. De-inking mechanisms

Chotipong et al. [40] proposed the mechanisms behind de-inking after a series of tests on CMC, wettability, zeta potential, solubilisation and de-inking performance. The proposed mechanisms are explained below and shown in Fig. 5.

Step 1: Surfactant adsorption or wetting on the printed and unprinted plastic surfaces. This process lowers the surface tensions between the ink-water and plastic-water interfaces. The cationic surfactant is composed of a hydrophilic head and hydrophobic tail groups. At the point of zero charge (PZC), it is expected that a stoichiometric pairing occurs between the positively charged surfactant head and negatively charged ink components. Due to the heterogeneity in composition, the ink surface may not be uniformly negatively charged. This results in a tail-in arrangement of the surfactant molecule across the plastic surface

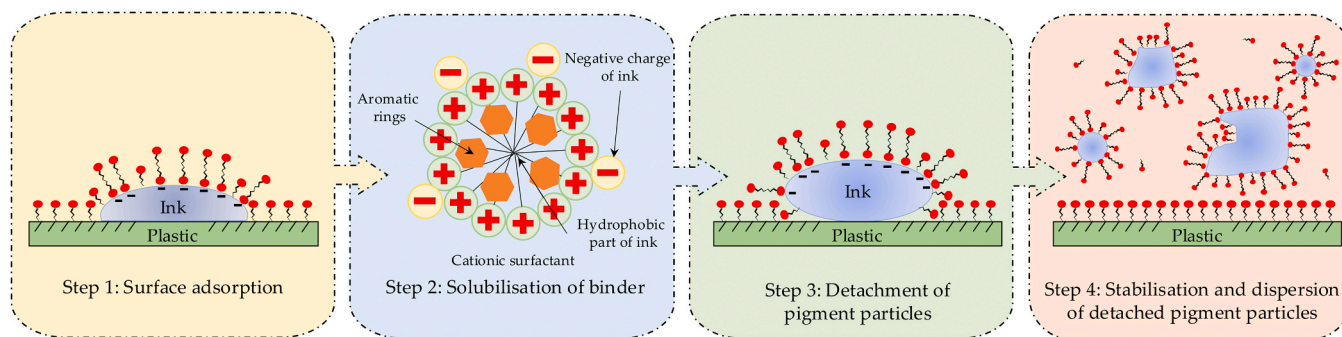


Fig. 5. Proposed four-step mechanism for surfactant-based de-inking. Adapted from Guo et al. [13].

(which is hydrophobic) and the positively charged areas on the ink surface. This tail-in arrangement is made possible by Van der Waal's force of interactions and results in a monolayer (hemimicelle) of surfactant molecules. A head-in tail-out arrangement of surfactant molecules is observed for the negatively charged areas on the ink surface. Progressively, this will result in a bilayer (admicelle) formation, with a second layer of surfactant molecules getting attached in a tail-in head-out arrangement over the previous layer. Complete wetting is achieved when the contact angle becomes 0° . However, it is difficult to differentiate between hemimicelle and admicelle formations, resulting in reduced contact angle and increased zeta-potential.

Step 2: Solubilisation of binder in micelles. This is hypothesised to be due to three factors: (i) the hydrophobic components of the ink being incorporated into the micelle core, (ii) electrostatic binding of the negatively charged carboxylate groups on the binder and the positively charged surfactant head groups on the micellar surface, (iii) the ion-dipole interactions between π -electrons of aromatic rings and hydrophilic surfactant head groups. Step (ii) is the most dominant de-inking mechanism, especially in cases where the surfactant concentration is above its CMC [13], and it happens simultaneously with Steps (i) and (iii).

Step 3: Detachment of ink from the plastic surface. This is hypothesised due to the electrostatic repulsion between the surfactant head groups. As the edge of the ink surface detaches, more surfactant molecules can be adsorbed to the newly uncovered areas on both the ink and plastic surfaces. It increases the electrostatic repulsion, resulting in further ink detaching.

Step 4: Stabilisation and dispersion of the detached ink particles. As more surfactant molecules are adsorbed to the detached ink particles and plastic surface, electrostatic stabilisation is achieved. Further, the electrostatic repulsion between the surfactant head groups on the plastic surface and detached ink prevents the redeposition of ink.

It can be noted that plastic de-inking is a surface reaction, which means the contact between ink on the plastic surface and the surfactant is vital to facilitate de-inking. This has been validated by Sánchez-Rivera et al. [39], who used a multi-layered printed plastic film and observed no significant de-inking compared to single-layer LDPE/HDPE films employed by several authors, including Gecol et al. [32], Songsiri et al. [38] and Chotipong et al. [40]. Despite varying surfactant

concentrations, reaction temperatures and pH were tested, including using a pressurised system, de-inking efficiency was poor with multi-layered printed plastic films.

Guo et al. [13] noted the presence of two different de-inking behaviours on single-layer printing ink systems, including solubilisation and peeling (Fig. 6). In solubilisation, the detached ink is observed to be dispersed in the bath, resulting in a cloudy solution with homogenous pigment colour. This was observed for NC and polyurethane (PU) binder systems (Fig. 6a). Meanwhile, for polyvinyl butyral (PVB) (Fig. 6b) and UV acrylic binder systems (Fig. 6c), the detached ink was found to have "peeled off" in tiny flakes and sediment in the water phase.

Binder systems can be classified into two based on the bonding mechanisms – (i) resulting in only Van der Waal's forces and hydrogen bonds due to polar groups in binders and the activated plastic surface from corona treatment, (ii) resulting in covalent bonds in addition to Van der Waal's forces and hydrogen bonds due to the presence of a reactive cross-linked system [13]. The latter enhances the bonding between the binder and the plastic surface, making it harder for ink removal. In such cases, a surfactant alone cannot cleave covalent bonds. In contrast, the surfactant hinders de-inking as it forms a protective layer over the ink surface when the hydrophilic-lipophilic balance (HLB) value is too low. Furthermore, alkaline hydrolysis can be an effective de-inking mechanism, as shown in Figure D.1 in Appendix D.

Based on the above discussion, a modified de-inking mechanism was proposed by Guo et al. [13], including (i) surfactant adsorption, (ii) breaking of hydrogen bonds via solubilisation and covalent bonds via chemical reaction, (iii) detachment of ink from the plastic surface through micelle solubilisation (for non-cross-linked binders) and mechanical peel off (for cross-linked binders), and (iv) stabilisation and dispersion of pigment particles.

5. Analysis of de-inking surfactants for packaging plastics

5.1. Cationic surfactants

Cationic surfactants are characterised by nitrogen-containing functional groups, such as amines, quaternary ammonium, imidazolium and pyridinium [41]. As the amine-based surfactants can only be protonated (activated) at acidic pH, their applications are limited to this pH range. Whereas quaternary ammonium-based surfactants ('quats') are not pH sensitive [42]. Cationic surfactants find major applications where surface adsorption is required (Fig. 7a), as most surfaces (including plastics) at neutral pH have a negative charge [43]. Thus, cationic surfactants are most explored for packaging plastic de-inking, even though they are the third by size in surfactant application. The various cationic surfactants employed in plastic de-inking are summarised in Table 2.

Cationic surfactants were effective in plastic de-inking above its CMC over a broad range of pH values. As cationic surfactants showed the most de-inking potential in early studies, the effect of various process parameters was explored in detail over time. Pre-soaking and mixing the

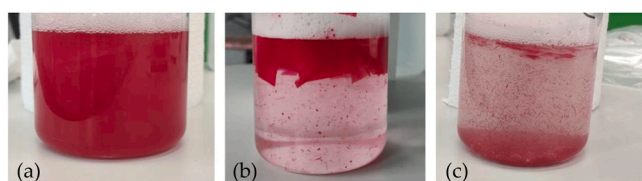


Fig. 6. (a) De-inking via solubilisation in NC binder, (b) De-inking via peeling in PVB binder, and (c) De-inking via peeling in UV-crosslinked binder [13].

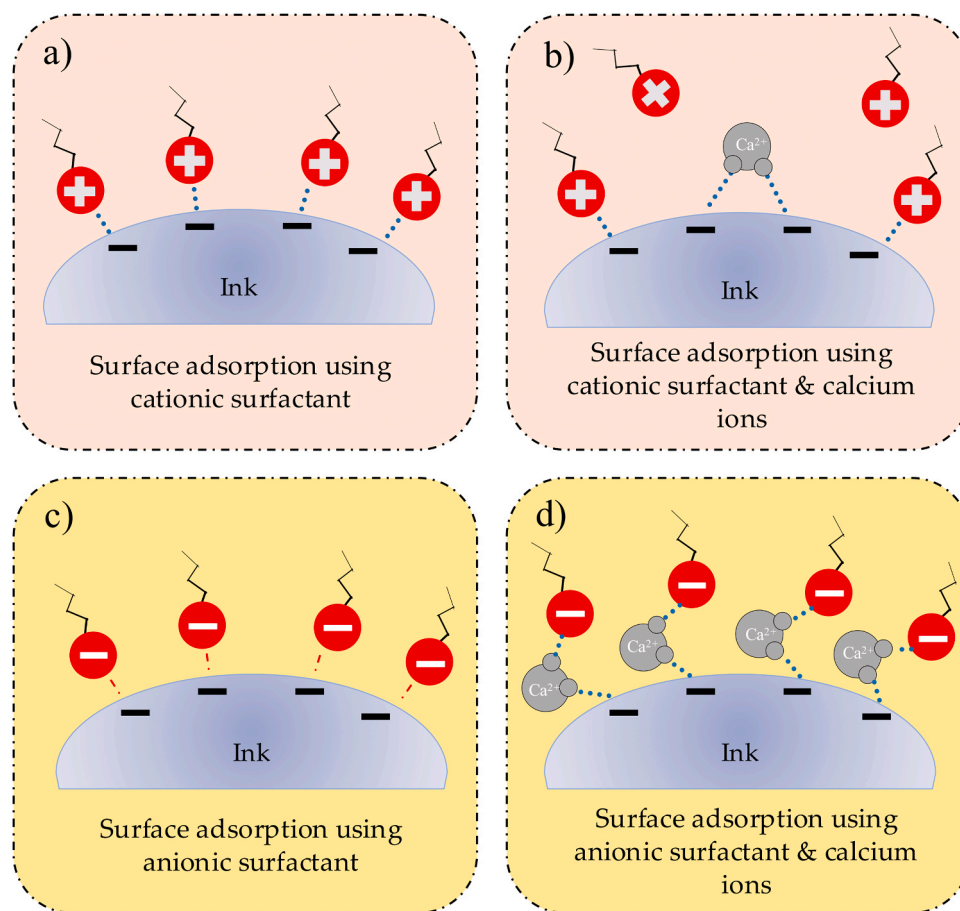


Fig. 7. Schematic diagrams depicting the surface adsorption of surfactants: a) cationic, b) cationic with calcium ions, c) anionic and d) anionic with calcium ions, on ink surface. Adapted from Gecol et al. [44].

samples via mechanical agitation improved de-inking efficiencies [38]. However, the design conditions are to be determined based on economic optimisations. More than 90% de-inking can be obtained with 5 h of pre-soaking and 2 h of mixing with hexadecyl trimethylammonium bromide at CMC. Conversely, if the pre-soaking time is reduced to 2 h, the mixing time increases to 4 h to obtain the same de-inking efficiency.

Increasing the lengths of alkyl chains was found to favour de-inking, as longer hydrophobes improved the adsorption rates [44]. The longer alkyl chains produced stronger Van der Waal's forces between the surfactant and ink particles [37]. Significantly high to complete de-inking can be achieved at highly alkaline pH. This is hypothesised due to the opening of epoxide rings (carrying negative charges) within the epoxy resins in inks, providing additional adsorption sites for the positively charged cationic head groups [37]. On the other hand, calcium ion addition was found to impede de-inking performance due to competitive adsorption (Fig. 7b).

5.2. Anionic surfactants

Anionics are the largest surfactant class used worldwide, accounting for roughly 70% of overall consumption [46]. This popularity is attributed to the ease and low cost of manufacturing. They are characterised by carboxylate, sulfonate, sulfate or phosphate functional groups [47] and are widely employed in detergent formulations and paper de-inking processes [48]. Anionics possess the highest cleansing power and good wetting properties. Table 3 summarises the anionic surfactants used in packaging plastic de-inking.

A few early studies in plastic de-inking explored the potential of

using anionic surfactants. Observations show that they are barely better than de-inking using water at respective pH. A possible explanation is the electrostatic repulsion between the negatively charged ink particles and the hydrophilic head of the surfactant, hindering adsorption (Fig. 7c). This is probably why recent studies have opted out of exploring anionic surfactants. However, adding calcium ions improved the de-inking capacity by providing additional sites for co-adsorption (Fig. 7d) [44]. Nevertheless, it was only found to be practical above the CMC of the surfactant in alkaline conditions.

5.3. Non-ionic surfactants

Non-ionics are the second largest class of surfactants used worldwide [41]. They are preferred due to the ease of production, readily available and inexpensive feedstock, and wide-ranging applications. This category is characterised by either polyether or polyhydroxyl functional groups [42]. A list of non-ionic surfactants used in packaging plastic de-inking is given in Table 4.

Early studies on non-ionic surfactants were conducted as they were less expensive and more biodegradable than amphoteric and cationic surfactants [49]. Generally, non-ionic surfactants are effective in plastic de-inking above its CMC in alkaline conditions. The exception is ethoxylated amines, which is a non-ionic. Despite that, it behaves like cationic in acidic conditions with low ethylene oxide levels [50]. Consequently, ethoxylated amines exhibit similar de-inking behaviour to cationic surfactants at acidic, neutral and basic pH, which is favourable. However, Guo et al. [12] observed no notable influence from pH values on de-inking efficiencies for non-ionic surfactants.

Table 2

Cationic surfactants used in packaging plastic de-inking.

Chemical	Observations	Reference
Dodecyl trimethylammonium bromide	Partial de-inking above CMC at slightly acidic, neutral and basic pH.	[44]
	100% de-inking at 24 times CMC at pH 12.	[40]
	At least 40% de-inking at 16.5 times CMC and pH 11.7. De-inking reduces with increasing temperature.	[37]
	Marginal improvements from NaCl addition.	
Hexadecyl pyridinium chloride	Effective between pH 4 –12 above its CMC.	[32]
Hexadecyl trimethylammonium bromide	Effective at slightly acidic, neutral and low basic pH above its CMC.	[32]
	Effective above CMC and at high pH (≥ 11.75). Can be improved by increasing temperature, pre-soaking time and mixing time (i.e. mechanical agitation).	[38]
	Effective above pH 11.	[36]
	Effective at slightly acidic, neutral and low basic pH above its CMC. Calcium addition reduces de-inking between pH 5 –12.	[44]
	100% de-inking at 4 times CMC at pH 12.	[40]
	At least 40% de-inking at 2 times CMC and pH 11.5. De-inking reduces with increasing temperature. Marginal improvements from NaCl addition.	[37]
	Over 90% de-inking using 1 wt% at pH 12.	[12]
	Over 90% de-inking for NC, PVB and UV-acrylate binders at selected conditions.	[13]
	No significant de-inking at various temperatures, mixing rates and pH in multi-layer plastics.	[39]
	Good de-inking medium for PUs. Exhibits a fast liquefaction process in cellulotics.	[11]
Modified waste cooking oil (WCOQE)	70% de-inking at pH 12 and 50 °C after 15 h.	[35]
Recycled PET-based surfactant (GT-Pyr-Br)	More than 90% de-inking at 0.5 times CMC and pH 1 –1.5.	[45]
Recycled PET-based surfactant (GD-Pyr-Br)	96% de-inking at CMC and pH 12.5.	
Tetradecyl trimethylammonium bromide	Effective at slightly acidic and slightly basic pH above its CMC.	[44]
	100% de-inking at 8 times CMC at pH 12.	[40]
	At least 40% de-inking at 6 times CMC and pH 11.8. De-inking reduces with increasing temperature. Significant improvements from NaCl addition.	[37]

Temperature promotes de-inking monotonically until the cloud point is passed, after which the efficiency reduces. In the cases of ethoxylated amines and ethoxylated alcohols, a lower cloud point resulted in higher de-inking efficiencies. Similarly, longer carbon chains in ethoxylated alcohols increased de-inking [49]. The addition of calcium ions was found to lower the de-inking efficiencies [44]. In the case of ethoxylated amines, this is explained by the fact that they behave as cationic surfactants at lower pH, creating a competition for adsorption sites on the ink particle.

5.4. Amphoteric surfactants

Amphoteric surfactants are uniquely equipped to exhibit anionic,

Table 3

Anionic surfactants used in packaging plastic de-inking.

Chemical	Observations	Reference
Sodium decylsulfate	Partial de-inking above CMC at basic pH (≥ 9).	[44]
Sodium dioctylsulfosuccinate	No significant de-inking using 1 wt% between pH 10 –12.	[36]
Sodium dodecanoate	Partial de-inking above CMC at basic pH (≥ 8). Calcium addition significantly improves de-inking.	[44]
Sodium dodecylsulfate	De-inking increases above CMC and between pH 8 –12. Yet only slightly better than de-inking with water.	[32]
	No significant de-inking using 1 wt% between pH 10 –12.	[36]
	Effective de-inking above CMC and high pH (≥ 10) by adding calcium	[44]
Sodium tetradecylsulfate	Effective de-inking above CMC and high pH (≥ 10).	[44]

Table 4

Non-ionic surfactants used in packaging plastic de-inking.

Chemical	Observations	Reference
Ethoxylated alcohols (AEO _x)	Effective de-inking above pH 10.	[49]
	Partial de-inking using 1 wt% at pH 12	[36]
	Calcium addition reduces de-inking between pH 7 –11, except for pH 12.	[44]
Ethoxylated amines (AMEO _{x+y})	De-inking effect increases with increasing acidic or basic pH.	[49]
	No significant de-inking using 1 wt % between pH 10 –12.	[36]
	Calcium addition reduces de-inking between pH 3 –12.	[44]
Oxirane, methyl-, polymer	90% de-inking using 1 wt% surfactant with 1 wt% macroemulsifier at pH 12.	[12]
	De-inking effect is generally low for various tested conditions. However, it softened the ink which can then be removed by friction.	[13]
Polyalkylene oxide modified polydimethylsiloxanes	Type with molecular weight 600 shows effective de-inking at pH 11.5 and 12.	[36]
Polydisperse nonylphenol polyethoxylated	100% de-inking observed at 25 mM (~510 times CMC) and pH 10.	[32]
	Effective above CMC at room temperature and pH 10.	[49]
Octoxynol 9	Over 90% de-inking for NC, PVB and UV-acrylate binders at selected conditions.	[13]

Table 5

Amphoteric surfactants used in packaging plastic de-inking.

Chemical	Observations	Reference
Dimethyl dodecylamine oxide	Below CMC de-inking increases at pH 11 and 12. Above CMC de-inking significantly improves without any trends to pH.	[32]
	No significant de-inking using 1 wt% between pH 10 –12.	[36]

cationic or zwitterionic properties. ‘Amphoterics’ are often confused with ‘zwitterionics’, though they are two distinct categories of surfactants [41]. In an amphoteric surfactant, the three states exist in equilibrium, depending on the pH range. Under acidic conditions, the

molecules will be protonated to form cations, whereas in an alkaline condition, they will be deprotonated to form anions. At the isoelectric point, they exhibit a zwitterionic character. Amphoteric surfactants are characterised by a secondary or tertiary amine group, which can be easily protonated [42]. The studies that employed amphoteric surfactants in plastic de-inking are listed in Table 5.

Over the years, only one amphoteric surfactant, dimethyl dodecylamine oxide, has been explored for plastic de-inking. Based on Gecol et al. [32], the amphoteric surfactant shows comparable de-inking capacities to cationic surfactant at pH 5, 7, 11 and 12 above its CMC. Whereas, below its CMC, a more consistent trend is observed with the anionic surfactant. Both the 'amphoteric' nature and the CMC of the surfactant depend on pH, and the effects of varying CMC on de-inking at various pH levels have not yet been explored.

6. Packaging plastic de-inking: the status quo

Packaging plastic de-inking studies are summarised in detail in Appendix D, and the effects of various parameters on de-inking efficiency are illustrated in Table 6.

Gecol et al. [32], the earliest work on packaging plastic de-inking, explored the potential of various surfactants (amphoteric, anionic, cationic, and non-ionic) in removing water-based inks from commercial plastic films. A control test was also designed using water to de-ink plastic films by varying the pH at room temperature. Results showed no significant de-inking until pH 10. At pH values 11 and 12, partial de-inking of 30% and 60% was observed. Several characterisation tests were conducted to understand the effect of inks on re-extruded plastic and how it impacts recycling. It was observed that there were significant colour changes, including the clear films turning beige and the printed films turning reddish brown. The decomposition of inks during re-extrusion released gases that caused crack points and holes across the cross-section. Ink aggregates were found to act as stress concentration points, reducing the strength of the material (e.g. modulus, ultimate strength and elongation at break are reduced). It was concluded that cationic surfactants are the most effective choice functional across various CMC and pH levels. Conversely, anionic surfactants were the least effective.

Following this, extended studies were conducted to understand various facets of de-inking. Exploring potential non-ionic surfactants, it was seen that nonylphenol polyethoxylate and ethoxylated alcohols exhibited good de-inking capacity at pH 10. In contrast, ethoxylated amines (which are amphoteric and classified as non-ionic but behave as cationic in acidic media) showed de-inking capacity at acidic, neutral, and basic pH [49]. Cationic surfactants, such as n-hexadecyl trimethyl ammonium bromide, exhibited de-inking capacity only above its CMC (complete de-inking at 5 times CMC) and high pH levels above 11.75.

In recent years, more systematic and non-conventional approaches to packaging plastic de-inking have been observed. Guo et al. [12], explored the possibility of using a new type of non-ionic surfactant oxirane, methyl-, polymer (commercial name: TEGO Surten W111) to de-ink LDPE film samples from post-consumer waste. In addition to the conventional CIELAB method for de-inking performance evaluation (Appendix C), a flake sorting method (Appendix C) is also considered to account for the heterogeneity of post-consumer plastic waste.

Guo et al. [13] explored the effects of various binders, pigments, over-print varnishes (OPV), cross-linkages in binders or OPVs and the materials' layering. Five detergents and four test conditions were included to analyse various possible scenarios. NC binder-based inks exhibited high de-inkability, and crosslinked binder systems can impede de-inking. Based on the above, it was concluded that breaking the Van der Waals forces, hydrogen bonds, and covalent bonds between inks and the plastic surface are critical steps in de-inking.

Üğdüler et al. [11] conducted an interesting study to understand the effect of seven different de-inking media in liquefying and solubilising 14 chemically different polymer resins in printing inks. The basic cost

Table 6

Relationships between de-inking parameters and efficiency.

Parameter	De-inking efficiency		Reference
Increasing temperature		Increases	[38]
		Increases until the cloud point of non-ionic surfactant	[49]
		Decreases due to the variations in the solubilisation capacity of ink binder in the surfactant	[37]
Increasing pre-soaking time	Increases		[38]
			[45]
			[49]
			[36]
Increasing mixing time	Increases		[38]
			[45]
			[49]
			[36]
Presence of abrasives	Increases		[38]
Increasing surfactant concentration	Significantly improves with cationic surfactants		[32]
			[38]
			[40]
pH	Acidic	Acid-based media were found to be more effective in liquefying the resins.	[11]
		Effective de-inking by cationic surfactants from recycled PET waste (GT-pyr-Br)	[45]
	Basic	High basic pH is required for effective and/or complete de-inking	[38]
			[36]
			[37]
		Surfactant-free basic solutions at high pH can attain partial de-inking for water-based ink	[32]
Salinity	Improves de-inking efficiency with alkyltrimethyl ammonium bromides (cationic surfactants)	Effective de-inking by cationic surfactants from recycled PET waste (GD-pyr-Br)	[45]
			[37]
Ink colour		Carbon black was the most challenging to remove	[36]
		Red ink can be removed by the addition of sodium silicate to cationic surfactants	
		1 wt% cationic surfactant at room temperature and pH 12 with pre-soaking removes all colours except black and red	
		Limonene (biosolvent), 1,2,3-propanetriol and 1,2,4-butanetriol (poly alcohols) resulted in 94% (blue) and 100% (orange) ink removal.	[51]
		No significant impact on de-inking, except in the case of carbon black	[13]
Addition of calcium ions	In amphoteric	Decreases	[44]
		Increases above the CMC at basic pH	
	In anionic	Increases for fatty acid soap (synthetic anionic surfactant)	
		In cationic	
	In non-ionic	Decreases	
Length of the surfactant hydrophobe	Increases		[49]
			[44]
			[40]
Improvements in rheological, thermal, and mechanical properties	No correlation to de-inking efficiency, though this contradicts the findings of Gecol et al.[32]		[12]

Table 7

Commercially available and patented technologies for packaging plastic de-inking.

Reference	Description
[52]	- Employed rotating brushes to de-ink unwound film rolls. - High de-inking efficiency of up to 99.9%. - However, the processed sample has poor mechanical/optical properties and odour. Hence not suitable for scaling up.
[53]	De-inking using alternating friction system and solvent based solution.
[54]	- Used to de-ink and de-odorise post-industrial polyolefin films. - Carbon dioxide is used in supercritical conditions. - Recycled films are suitable to be used in secondary packaging applications.
[55]	- De-inking using water-based solutions without any environmentally harmful chemicals. - High quality recyclates are obtained, and those can be utilised for high-end value-added product manufacturing. - The system also includes a design to minimise water consumption and recover the de-inking chemicals.
[56]	- Similar design as [52]. - De-inking is limited to plastic film rolls. - In order to process large quantities a grinding step is added.
[57]	- De-inking and de-lamination using oxidising inorganic acids and fatty acids. - Suitable for water-based, solvent-based and UV-based inks

assessment also determined that the price of the de-inking agent and its required volume played a crucial role in the economic feasibility of the de-inking process.

Only four studies attempted sustainably sourced solvents in plastic de-inking by utilising biomass-derived or valorised waste materials in surfactant production. El-Sherbiny et al. [45] developed novel cationic surfactants from recycled PET waste (GT-pyr-Br and GD-pyr-Br – Table 6), which were achieved by depolymerising PET followed by esterification and quaternization. Gd-pyr-Br resulted in a maximum de-inking efficiency of 96% at CMC. Ferreira et al. [51] performed solvent extraction using the dissolution-precipitation technique to remove blue and orange pigments from HDPE using a range of bio-solvents. 94% and 100% de-inking were observed for blue and orange pigments,

respectively. In this method, a solvent is initially used to solubilise HDPE and then an anti-solvent is used to selectively precipitate the polymer. Both the solvent and anti-solvents could be recovered and reused, ensuring economic and environmental sustainability. Ye et al. [35] explored the potential of quarternary ammonium-modified waste cooking oil (soybean oil) in de-inking. This was prepared by ring-open esterification and quaternary amination reactions. De-inking efficiency of around 70% was observed through this method. Sánchez-Rivera et al. [39] reported a solvent-targeted recovery and precipitation (STRAP) strategy to de-ink a multilayer plastic film from post-industrial sources. It was seen that biomass-derived gamma-valerolactone (GVL) was able to de-ink all the studied colours. The techno-economic analysis also showed that the STRAP process could be economically viable for a processing capacity of 6000 tonnes per year. This shows the efficacy of bio-based and waste-based surfactants, and the de-inking efficiencies are comparable to those of synthetic ones.

Considering the potential industrial applications of packaging plastic de-inking, a few commercially available and patented technologies are listed in Table 7 for a comprehensive overview.

7. Impacts of conventional de-inking surfactants

Drawing parallels with the detergent laundry processes, surfactants are predominantly used in packaging plastic de-inking. However, the use of surfactants poses several ecotoxicological and health hazards (Fig. 8). Anionic surfactants, such as linear alkylbenzene sulfonates (LAS), cause pathological, biochemical, and physiological impacts on terrestrial and aquatic ecosystems. It also interferes with marine biota's environmental stress, reproduction, and growth. Surfactants facilitate eutrophication by increasing the solubility of contaminants in water bodies (Fig. 8). Their toxicity increased proportionally with increasing hydrophobic properties [58].

Even wastewater treatment plants cannot properly manage the surfactants, which are incompletely degraded and discharged into water bodies. Surfactant concentrations above 0.1 mg/L in water bodies cause huge, persistent foams that prevent oxygen adsorption [59]. It also impacts the hydrological cycle by impeding evaporation. When the surfactant concentration increases beyond the CMC, higher concentrations of soluble and insoluble pollutants in the water are observed [59].

Surfactants also interfere with soil properties, primarily by reducing the capillary pressure at the air-water interface [60]. This causes an

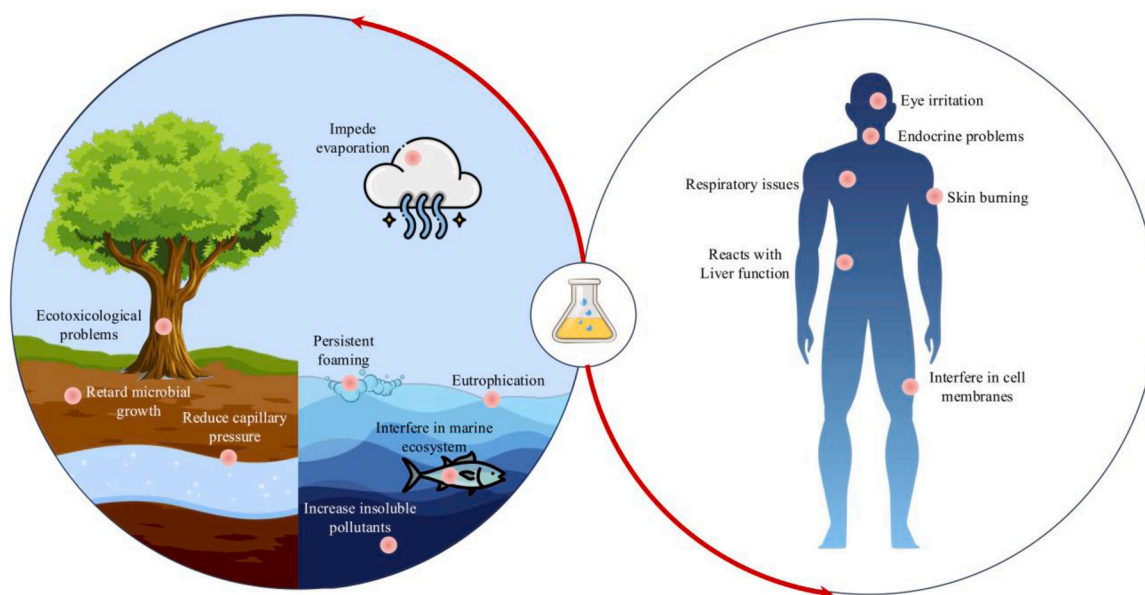


Fig. 8. Potential environmental and human health hazards from surfactant use.

altered air and water distribution in the soil's pore system and reduces water retention and infiltration (Fig. 8). Irrigating soil with surfactant-rich water increased the soil salinity, electrical conductivity, sorptivity and pH, while hydraulic conductivity was reduced [61]. It was also studied that surfactants retarded microbial growth and increased microbial mutation and mortality [62].

In humans, when ingested, surfactants react with proteins in the liver and serum, causing long-term metabolic effects and disrupting the endocrine system [63]. It was reported to cause skin burning, eye irritation and respiratory problems. Nonylphenol ethoxylates were studied to have estrogenic effects in mammals, Pisces, and amphibians [64] (Fig. 8). Surfactants, such as LAS, can change the cell membrane structure and impair the transpiration and translocation of water and essential nutrients [58]. Sodium dodecyl sulfate inhibits ATPase activity, damages membrane structures and initiates oxidative stress [65].

In an attempt to reduce the impacts of plastic waste, it is vital not to unintentionally cause secondary impacts through the use of hazardous and toxic chemicals and surfactants. Thus, the surfactant selection in packaging plastic de-inking applications is paramount.

8. Modelled AHQ values for chemical selection

Calculated AHQ values based on the GHS can be used for one-to-one comparison of chemical hazards (Table 8). Based on the current dataset, cyclohexane would be the most hazardous chemical (check the note in Appendix B). Cationic surfactants (e.g. hexadecyltrimethylammonium bromide, cetyltrimethylammonium chloride, dodecyltrimethylammonium bromide, hexadecylpyridinium chloride) have been extensively studied for de-inking, though possess the maximum environmental AHQ. Hexadecyltrimethylammonium bromide, explored in 9 studies, has a concerning total AHQ of 1.43 out of the maximum possible 3. In this way, informed decisions can be made when selecting appropriate solvents for de-inking purposes.

All chemicals with a total AHQ less than 0.5 in this work match with the signal words “warning” or “nothing” in the SDS. The only exception is polydisperse nonylphenol polyethoxylate with the signal word “warning”, though the calculated AHQ is 1.0. In short, the consistent method employed in work to ascertain AHQ values is satisfactory, though improvements can be made based on the significance level of hazard classes (Eq. 1 and check the note in Appendix B).

Appropriate chemical selection tools, such as AHQ, should be utilised to minimise potential hazards and toxicities from plastic waste recycling. While AHQ is a preliminary chemical assessment tool, local socio-techno-economic factors, such as low chemical cost, desirable chemical action, low carbon footprint and local availability, should also be considered to improve the quantitative assessment of chemical and surfactant selection.

9. Future directions

- Material characterisation tests should be conducted to address the conflict of opinion on whether de-inking improves recyclate quality.
- Only Guo et al. [13] explored the impacts of ink pigments, binders, OPVs and cross-linking in de-inking. Future studies can characterise the most common ink types, categorise them based on application (food contact materials and others) and identify whether a specific de-inking solution is required for each application.
- Carbon black was identified as one of the most challenging inks to remove [13,36]. Previously, Gecol et al. [36] demonstrated that a mixture of cationic and non-ionic surfactants can effectively de-ink carbon black. However, this was not studied further. Future studies can explore the performance of this mixture for other ink-substrate combinations and whether it can be utilised for post-consumer plastic waste.
- The potential for solvent reuse should be explored further to ensure sustainability. Only Ferreira et al. [51] demonstrated that their

Table 8

Calculated AHQ values for various de-inking chemicals employed in the literature. The bold values indicate the maximum values (i.e. 1) in each category. Note: A sample calculation is given in Appendix B. Some SDS indicate 0 physical, health or environmental hazards for some chemicals (e.g. 1-Butanol), and it does not necessarily imply the chemical is hazard-free.

Name	Average Hazard Quotient			
	Physical	Health	Environment	Total
1-Butanol	0.11	0.36	0.00	0.47
1-Hexanol	0.11	0.21	0.00	0.32
1-Pentanol	0.11	0.42	0.00	0.53
1-Propanol	0.50	0.56	0.00	1.06
1,2-Butanediol	0.00	0.50	0.00	0.50
1,2-Ethanedil	0.00	0.28	0.00	0.28
1,2-Hexanediol	0.00	0.50	0.00	0.50
1,2-Pentanediol	0.00	0.50	0.00	0.50
1,2-Propanediol	0.00	0.00	0.00	0.00
1,2,3-Propanetriol	0.00	0.00	0.00	0.00
1,2,4-Butanetriol	0.00	0.37	0.00	0.37
1,3-Butanediol	0.00	0.50	0.00	0.50
1,3-Propanediol	0.00	0.00	0.00	0.00
1,4-Butanediol	0.00	0.09	0.00	0.09
1,5-Pentanediol	0.00	0.00	0.00	0.00
1,6-Hexanediol	0.00	0.00	0.00	0.00
2-Butanol	0.11	0.24	0.00	0.35
2-Propanol	0.50	0.31	0.00	0.81
Acetone	0.50	0.31	0.00	0.81
Cetyltrimethylammonium chloride	0.00	1.00	1.00	2.00
Cyclohexane	0.50	0.54	1.00	2.04
Dimethyl dodecylamine oxide	0.00	1.00	0.00	1.00
Dodecyltrimethylammonium bromide	0.00	0.37	1.00	1.37
Ethanol	0.50	0.50	0.00	1.00
Ethyl acetate	0.50	0.31	0.00	0.81
Formic Acid	0.11	0.54	0.00	0.65
Hexadecylpyridinium chloride	0.00	0.43	1.00	1.43
Hexadecyltrimethylammonium bromide	0.00	0.43	1.00	1.43
Limonene	0.11	0.83	1.00	1.94
Methanol	0.50	0.33	0.00	0.83
N,N-Dimethylcyclohexylamine	0.11	0.53	1.00	1.65
Sodium Hydroxide solution	1.00	1.00	0.00	2.00
Oxirane, methyl-, polymer	0.00	0.00	0.11	0.11
Polyalkylene oxide modified polydimethylsiloxanes (600)	0.00	0.22	0.00	0.22
Polyalkylene oxide modified polydimethylsiloxanes (1000)	0.00	0.28	0.00	0.28
Polydisperse nonylphenol polyethoxylate	0.00	0.50	0.50	1.00
Polyethylene glycol <i>tert</i> -octylphenyl ether	0.00	0.52	1.00	1.52
Sodium dioctylsulfosuccinate	0.00	0.75	0.00	0.75
Sodium dodecanoate	0.00	0.54	0.00	0.54
Sodium Dodecylsulfate	0.50	0.35	0.00	0.85
Sodium tetradecyl sulfate	0.00	0.37	0.00	0.37
Sulfuric acid	1.00	1.00	0.00	2.00
Tetradecyltrimethylammonium bromide	0.00	0.43	1.00	1.43
Toluene	0.50	0.52	0.00	1.02
Xylene	0.11	0.40	0.00	0.52
Maximum AHQ values in this table	1.00	1.00	1.00	2.04

solvents are reusable for up to three cycles. The life cycle of the de-inking surfactant or chemical should also be adequately evaluated to understand the end-of-life disposal and associated impacts. Aligning with the principles of circular economy, alternative bio-based surfactants can be explored, as done by El-Sherbiny et al. [45], Ferreira et al. [51], Sánchez-Rivera et al. [39] and Ye et al. [35].

- Bioaccumulation of de-inking chemicals and their related health and environmental impacts should be thoroughly studied through pilot plants and industrial applications.

- The majority of the studies have focussed on uni-coloured plastic films. In real life, printed materials utilise several colours. Thus, treatments to achieve the desired quality and output with multi coloured printed materials are required.
- Mixed and contaminated plastic streams from post-consumer applications are available. Hence, the applicability of the identified surfactants in de-inking this complex post-consumer plastic waste should be thoroughly explored.
- The recycling industry should investigate the feasibility of integrating conventional mechanical and chemical recycling methods and supplementing them with the studied de-inking processes.
- Techno-economic and life cycle assessment (LCA) studies should also be conducted to understand the feasibility and impacts of commercialising de-inking processes in the recycling industry.

10. Conclusions

With 175 nations agreeing to develop a legally binding Global Plastics Treaty by 2024, all the aspects of plastic recycling should be thoroughly explored. Obtaining the highest quality of recyclates will be paramount in moving towards a circular plastics economy. While advancements have been made across various fronts in plastic recycling, de-inking was found to be an undermined pre-treatment step. This study provided a critical understanding of plastic de-inking's state of the art. While surfactants are conventionally utilised in de-inking owing to similarities with laundering processes, a general caution needs to be exercised regarding potential hazards and toxicities. The proposed AHQ model could be an effective chemical rating and selection tool. Advancements in plastic de-inking will contribute towards the United Nations Sustainable Development Goals, such as Industry, innovation, and infrastructure (Goal 9), Responsible consumption and production (Goal 12), and Climate Action (Goal 13).

Environmental implication

Packaging plastic waste constitutes over 40% of total plastic waste but is often neglected in recycling due to contamination and complex separation needs. Effective de-inking can significantly enhance recycle

quality by removing contaminants, such as printing ink. This study critically evaluates current de-inking methods and introduces the Average Hazard Quotient (AHQ) to assess the toxicity and hazards of de-inking chemicals. By minimising environmental impacts and reducing toxic chemical release, the research supports cleaner recycling practices and promotes the development of greener alternatives. Ultimately, it advances plastic circularity by facilitating the production of higher-quality recyclates from packaging plastics.

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CRediT authorship contribution statement

Shafeeq Ahmed Syed Ali: Writing – original draft, Validation, Methodology, Data curation, Conceptualization. **I.M.S.K. Ilankoon:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Lian Zhang:** Writing – review & editing, Resources. **Jully Tan:** Writing – review & editing, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

A. Additional discussion on bibliometric study

The annual scientific production is an appropriate measure to evaluate a research field's emerging or declining significance. Figure A.1 shows that there has been an increased interest in research related to packaging plastic de-inking.

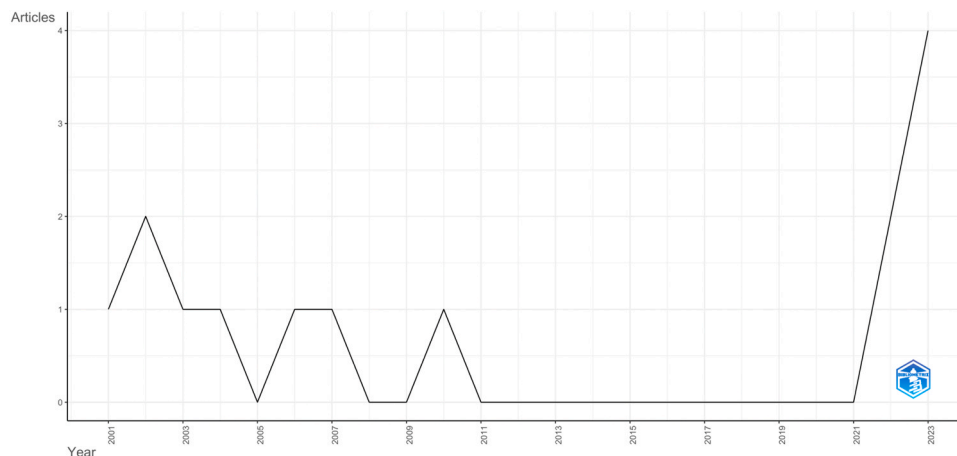


Figure A.1. Annual scientific production of packaging plastic de-inking literature.

In a thematic map, the Cartesian coordinates characterise the research topics based on centrality and density. This helps to classify the topics into motor themes, niche themes, emerging or declining themes and basic themes. Motor themes possess high density and centrality and act as the driving force of the research field. In this case, as seen in Figure A.2, the overlap between the critical topics, such as ‘ink’ and ‘polyethylene’, which are commonly used in packaging plastic applications, are the driving forces. On the other hand, with low centrality and low density, emerging or declining themes represent marginal or underdeveloped fields. CMC, pH effects and adsorption fall into the basic themes, with high centrality and low density. This reflects cross-cutting topics that are well connected to the rest yet poorly developed themselves. Finally, niche themes, including plastic packaging, dissolution and environmental impact, have been explored thoroughly. Yet they are poorly connected with the rest of the field.

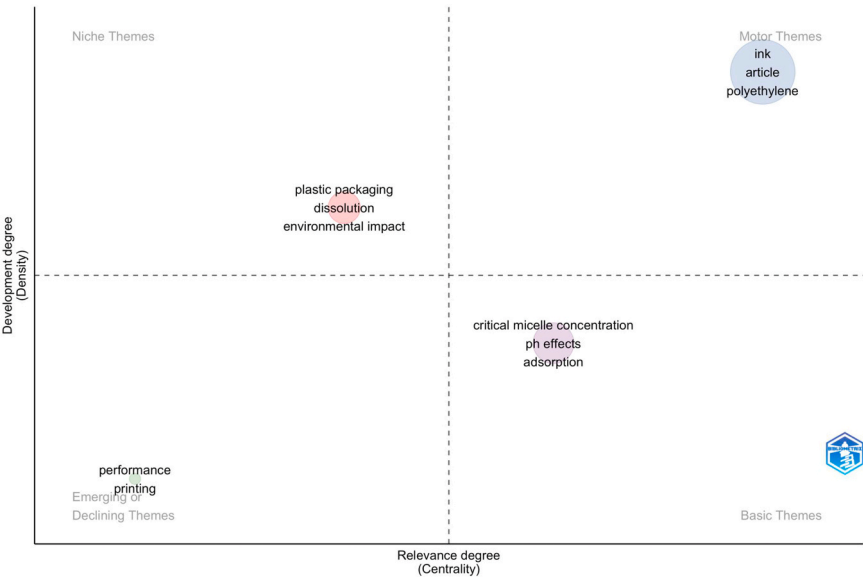


Figure A.2. Thematic map for packaging plastic de-inking studies.

Over time, when a pair of publications are cited together, they form a research cluster. Each of these clusters then follows a common theme. Co-citation analysis is, hence, an interesting way to schematically represent the thematic evolution of the field. The co-citation network for packaging plastic de-inking literature is presented in Figure A.3. Furthermore, keyword analysis helps to identify the popular research themes and trends in the field. The treemap for common keywords in de-inking is presented in Figure A.4. The top five keywords are ink, recycling, article, de-inking, and polyethylene.

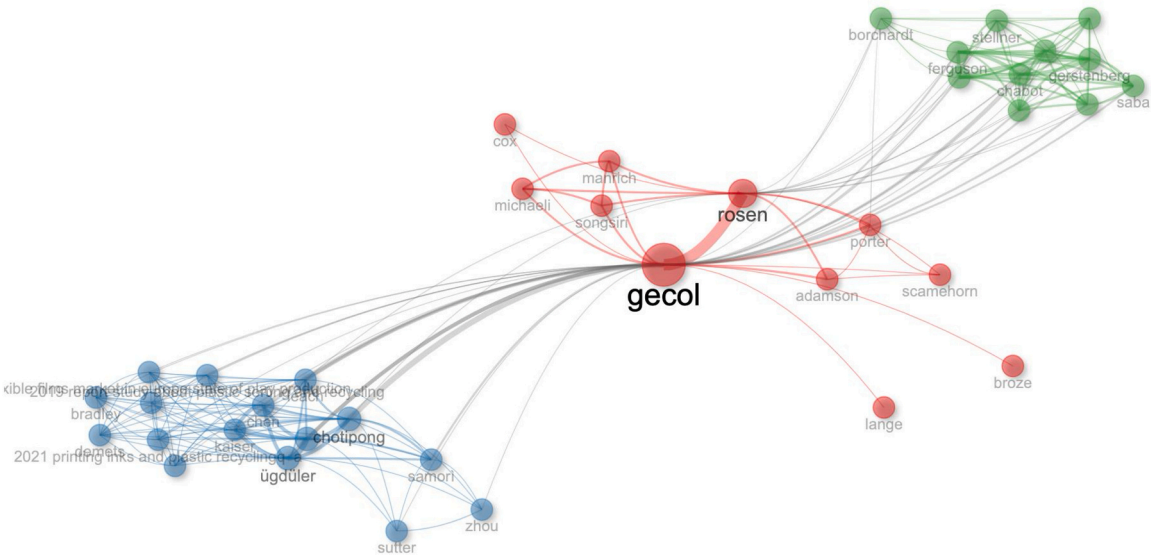




Figure A.4. Treemap highlighting the popular keywords in the de-inking literature.

B. Sample AHQ calculation

The GHS data for cyclohexane was collected from its SDS, as shown in Table B.1.

Table B.1
GHS-based hazard classification data for cyclohexane from SDS.

Name	Physical		Health		Environment	
	Code	Category	Code	Category	Code	Category
Cyclohexane	H225	2	H315	2	H400	1
			H336	3	H410	1
			H304	1		

A sample AHQ calculation using Eq. (1) is shown below:

$$AHQ = 1 \times \left(\frac{1}{1} \times \frac{1}{2^1}\right) + 1 \times \left(\frac{1}{3} \times \left(\frac{1}{2^1} + \frac{1}{3^2} + \frac{1}{1^1}\right)\right) + 1 \times \left(\frac{1}{2} \times \left(\frac{1}{1^1} + \frac{1}{1^1}\right)\right)$$
$$AHQ = 0.5 + 0.54 + 1 = 2.04$$

This method limits the AHQ value for each hazard class (physical, health and environmental) between 0 and 1, whereas the total AHQ value is limited between 0 and 3. The AHQ values of 45 chemicals used in literature for de-inking were calculated similarly and presented in Table 8.

Note: In the above calculation, $r = 1$. However, suppose it is expected that there is a higher probability of a particular hazard. In that case, the r value can be increased based on the significance level for that hazard class. For example, $r_{\text{physical}} = 0.9$; $r_{\text{health}} = 1.5$; $r_{\text{environmental}} = 0.6$. In this case, significant health hazards are expected, or the researchers are more interested in selecting a de-inking surfactant with lower health hazards. This will result in an overall AHQ of 1.86 for cyclohexane. Hence, the choice of significance level or r value is crucial in calculating the AHQ.

C. Methods used in de-inking performance evaluation

C.1. CIELAB method (by the International Commission on Illumination)

Eight of the fourteen studies in de-inking literature have employed a standardised evaluation, the CIELAB method. It was first devised by the International Commission on Illumination in 1976 based on the tristimulus values of a colour. It quantifies the red, green, and blue light required to produce that colour [66]. A spectrophotometer or a colorimeter is used to evaluate the CIELAB values (L^* , a^* , b^*) (Figure C.1).

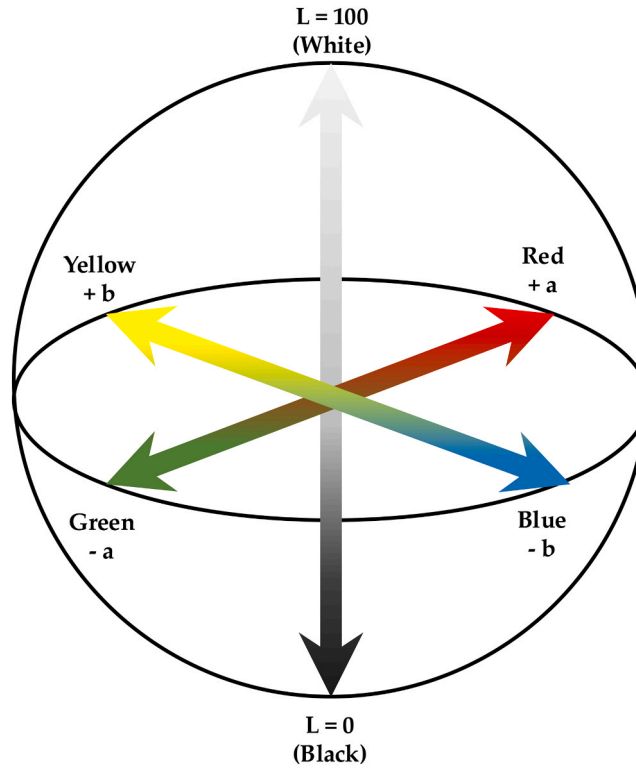


Figure C.1. Schematic of the three-dimensional CIELAB colour model (adapted and redrawn from [67]).

Once the averaged values for L^* , a^* and b^* are obtained, the de-inking efficiency can be calculated as follows:

$$DE^* = \frac{\sqrt{\Delta L_{\text{sample-blank}}^2 + \Delta a_{\text{sample-blank}}^2 + \Delta b_{\text{sample-blank}}^2}}{DE_{\text{blank}}^*} \quad (\text{C.1})$$

where, DE^* - De-inking efficiency. DE_{blank}^* - De-inking efficiency for reference material. ΔL , Δa , Δb - Differences in CIELAB values between the sample and reference materials.

C.2. Flake sorting method

To account for the heterogeneity of post-consumer plastic waste, Guo et al. [12] implemented a modified approach to analysing the de-inking efficiency. The de-inked samples were manually sorted into – fully transparent, partially transparent, and opaque (non-de-inked). The weight of each of these sample groups is measured, and the de-inking efficiency is then calculated as follows:

$$DE^* = \frac{(m_{ct} + m_{pt} \times 0.5)}{m_{sample}} \quad (C.2)$$

where, DE^* - De-inking efficiency. m_{ct} - The mass of fully transparent flakes (in grams). m_{pt} - The mass of partially transparent flakes (in grams). m_{sample} - The mass of the entire selected sample (in grams).

Guo et al. [12] showed significant variations between the CIELAB and the flake sorting methods. It was determined that the CIELAB method displayed a high standard deviation (11% to 76%) and was suggested not to be the optimal choice for evaluating post-consumer plastic waste. However, CIELAB is still the most suitable method for a homogenous sample.

C.3. Scanning and pixel reading

Another approach observed in the literature is scanning the plastic specimen directly or sticking it to a white paper using transparent glue [40,37, 38]. The scanned copies were then analysed using software, such as Adobe Photoshop, to count the number of pixels (in proportion to the amount of ink) on the plastic surface. The de-inking efficiency is then calculated as follows:

$$DE^* = \frac{(Pixels_{before\ de-inking} - Pixels_{after\ de-inking})}{Pixels_{before\ de-inking}} \times 100 \quad (C.3)$$

Meanwhile, Guo et al. [13] developed a Python code to read the pixels from the scanned files and generate the L^* , a^* , and b^* values as per the CIELAB standards. The output from running the Python can then be used in Equation C.3 to calculate the deinking efficiency. Another Python code was developed by Ügdüler et al. [11] using the Open-Source Computer Vision (OpenCV) Library. The scanned images were first grey-scaled and processed to obtain arrays with values ranging between 0 (black) to 255 (white). This represented the shades of grey and corresponded to the amount of ink remaining on the plastic film. The de-inking efficiency can then be calculated as follows:

$$DE^* = \frac{(d_{sample} - d_{min})}{(d_{max} - d_{min})} \times 100 \quad (C.4)$$

Where, d_{sample} - the average grey value of the sample. d_{max} - the average grey value of the fully printed plastic film. d_{min} - the average grey value of the transparent plastic film.

D. Literature review on packaging plastic de-inking studies

The schematic for the modified de-inking mechanisms incorporating alkaline hydrolysis is shown in Figure D.1. A detailed summary of de-inking literature related to packaging plastics is provided in Table D.1.

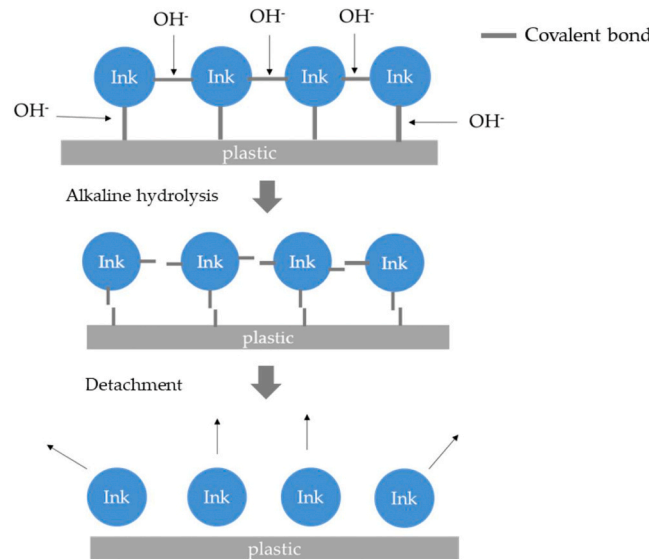


Figure D.1. The proposed de-inking mechanism through alkaline hydrolysis for crosslinked binder systems [13].

Table D.1

Summary of literature review on packaging plastic de-inking.

Source	Plastic type	Ink type	Tests conducted	Parameters evaluated	Observations
[32]	Commercial PE film	Water-based	SEM, TEM, EDX, Tensile test, Spectrocolorimeter	CMC, pH	- Re-extrusion of PE films resulted in colour change: clear film became beige, and the printed film became reddish brown. - Ink contamination results in a weaker and less stiff plastic upon re-extrusion. - The re-extruded pellets displayed crack points and holes across the cross-section. - Cationic surfactants are most effective in de-inking. Possibly due to the acidic acrylate in water-based ink. - Amphoteric surfactants are only effective above their CMC over a range of pH. - Nonionic surfactants are only effective above its CMC at pH 10. - Anionic surfactants are barely better than water even above CMC and high pH. - Complete de-inking with CTAB is observed above 5 times CMC, pH 12, 30 °C, 2 h soaking and 2 h mixing time. - Abrasives speed up the de-inking process, however it may not be strictly required.
[38]	Rigid HDPE	Solvent-based	Scanjet scanner	Concentration, pH, temperature, pre-soaking time, mixing time and abrasion	- Pre-soaking and mixing time positively impact de-inking. 10–80% ink removal can be obtained by increased pre-soaking time from 0 to 3 h. - Economic optimisations may be attempted to determine the optimal pre-soaking and mixing time. - pH above 11.5 is needed to display the de-inking effect. - Temperature and de-inking have a positive correlation. - NP(EO)₁₀ is only effective above its CMC at room temperature and pH 10.
[49]	Commercial PE film	Water-based	Spectrocolorimeter, TEM, EDX, Tensile test, Differential Scanning Calorimetry, Density gradient test	pH, agitation time, pre-soaking, temperature, HLB, size of PE film	- AEO₅ exhibited comparable de-inking properties to NP(EO)₁₀, which was increasing with respect to increasing basic pH. Most effective at pH 10 or above. - AMEO₅ performs de-inking over a wide range of pH. Residual ink on the PE film reduces proportionally to increasing acidity or alkalinity. - Between 0 to 180 min agitation has a positive impact on de-inking, after which it stabilises. - Pre-soaking for 24 h significantly improved the de-inking performance. - Between room temperature to 60 °C, de-inking increases with increasing temperature for NP(EO)₁₀. - However, as the temperature is increased to 75 °C, the de-inking reduces. This is because the temperature is higher than the cloud point of the surfactant. - De-inking increased as the HLB of the surfactant reduced (decreasing the number of EO_x groups) for both ethoxylated alcohols and amines, except in the cases of AEO₃ and AMEO₂. - Reducing the width of PE film strips to 12.7 mm from 25.4 mm significantly improved de-inking and reduced the error. However further reducing the width to 6.4 mm from 12.7 mm did not exhibit a significant improvement. - The ink aggregates that degrade the plastic quality on re-extrusion disappeared when treated with AEO₅ and AMEO₅. - No de-inking is observed when surfactant free basic solution is used between pH 3–12. - Cationic surfactant is determined to be the most effective in solvent-based ink removal.
[36]	Commercial PE film and Uni coloured PE films	Solvent-based	SEM, EDX, Spectrocolorimeter	Concentration, pH, agitation time, pre-soaking	- Removal of black ink (carbon black) was identified to be a significant challenge in de-inking. - Complete removal of all colours, except red and black, can be obtained using 1 wt% cationic surfactant at room temperature and pH 12 with pre-soaking.

(continued on next page)

Table D.1 (continued)

Source	Plastic type	Ink type	Tests conducted	Parameters evaluated	Observations
[44]	Commercial PE film	Water-based	Spectrocolorimeter, Zeta-potential	Addition of hardness ions (calcium ions), length of surfactant hydrophobes	- Addition of sodium silicate to the cationic surfactant completely removes red ink. - De-inking of solvent-based inks requires extreme conditions and a selective choice of surfactants in comparison to water-based inks. - Addition of calcium ions to fatty acid soap (synthetic anionic surfactant) at alkaline pH was nearly as effective as cationic, nonionic, or amphoteric surfactants alone. - However, the addition of calcium ions reduced the de-inking capacity of cationic, nonionic and amphoteric surfactants. - Length of the surfactant hydrophobe was seen to have a positive correlation with de-inking capacity. - De-inking efficiency increased in relation to increasing alkyl chain length or surfactant concentration. - Complete de-inking was observed at concentrations higher than the CMC for CTAB (3 times), TTAB (8 times) and DTAB (24 times).
[40]	HDPE bottles	Solvent-based	CMC, Wettability, Zeta-potential, Viscosity, Solubilisation	CMC, length of alkyl chains, Wettability, Zeta-potential	- It was observed that as contact angle reduces (wettability increases), solubilisation of ink particles increases and zeta-potential increases with increasing surfactant concentration or alkyl chain length. - CMC of the surfactants increased slightly with increasing temperature. However, it decreased significantly by adding 0.1 M NaCl. - Zeta-potential became more positive (or surfactants adsorbed to the ink particles) more easily with increasing pH, concentration or alkyl chain length.
[37]	HDPE bottles	Solvent-based	Zeta-potential, CMC, Solubilisation	pH, temperature, salinity	- Solubilisation of ink binder was seen to increase with decreasing concentration, increasing temperatures and increasing salinity. - Salt addition was also seen to significantly increase the complete de-inking CMC value for CTAB and TTAB. - De-inking was found to increase with decreasing temperature, increasing pH, increasing temperature and increasing salinity. - New cationic surfactants were synthesised from PET bottles through depolymerisation, esterification, and quaternisation. - GD-Pyr-Br was only effective in de-inking at pH 12.5 and 13. - Whereas GT-Pyr-Br was only effective between pH 1 to 3, where the de-inking capacity is reduced with increasing pH. - For GD-Pyr-Br, the de-inking was proportional to surfactant concentration, reaching 96% at CMC. - More than 90% de-inking can be attained above 0.5 CMC, at pH 1 to 1.5. - Pre-soaking for up to 1 h improved the de-inking for GD-Pyr-Br for all concentrations, after which the de-inking increased up to 2 h in case of concentrations 0.5 CMC and below. Whereas for concentration at CMC and above, the de-inking significantly reduced. - In the case of GT-Pyr-Br, pre-soaking improved de-inking for all concentrations. - Significant improvements in the de-inking effect were observed for both surfactants when mixing time increased up to 1 h.
[45]	LDPE film	Solvent-based	Molecular weights, surface tension, colour reflection densitometer, spectrophotometer, FTIR, H NMR spectroscopy	Concentrations, pH, soaking time, mixing time	- As a single component system CNF had the least impact on water turbidity removal and de-inking. - As a single component system cPAM was effective in removing 18.5% of water turbidity and 68.2% of yellow ink. However, it has an insignificant impact on blue and black inks. - The dual component system of CNF followed by cPAM successfully de-inked 94% of blue, 87.5% of black and 90% of yellow inks. Water turbidity removal also significantly improved in this case.
[68]*	-	Water-based	Focused Beam reflectance measurement (FBRM)	flocculation	

(continued on next page)

Table D.1 (continued)

Source	Plastic type	Ink type	Tests conducted	Parameters evaluated	Observations
[51]	HDPE		EDXRF, Raman microscopy, FTIR, TGA, DSC, Optical microscopy	Alkyl chain length, number of hydroxyl groups, the distance between -OH groups in the alkyl chain,	• Optimum condition identified at 0.01 wt% CNF and, 0.0075 wt% (blue) and 0.01 wt% (yellow and black) cPAM dose. • Flocculation studies show that the ink is first adsorbed onto the CNF. After which it flocculates due to cPAM addition. • A dissolution-precipitation technique is employed. • Increasing the number of -OH groups in the alcohol-favoured pigment removal. • While increasing the number of carbons in the alkyl chain, pigment removal increases until 4 carbons, after which it reduces due to a decrease in polarity. • The pigment removal was most effective for tri-alcohols. • The most feasible solvent/antisolvent combination was identified as limonene/1,2,3-propanetriol (glycerol). • New surfactant is synthesised from waste cooking oil using ring-opening esterification and quaternary amination reaction. • WCOEQ had higher foam stability and foaming ability in comparison to SDS and CTAB. These are important parameters for better de-inking. • WCOEQ had a lower kraft point than SDS (less than 0 °C) which presents a wider application. • In comparison to traditional cationic surfactants, WCOEQ has stronger emulsifying ability and higher adsorption strength. • WCOEQ exhibited optimum de-inking performance between 5 to 7 pH. • Optimum de-inking was observed with a pre-soaking of 4 –8 h, stirring of 1 –2 h, and temperature at 50 °C. • CTAB had the best de-inking effect (94.2%) at pH 12 based on both CIELAB value and flake sorting method. • The non-ionic surfactant only showed up to 65% de-inking at an alkaline pH of 11 –12, using the CIELAB method. • The non-ionic surfactant added with a macro-emulsifier showed similar de-inking performance (90% at pH 12) to CTAB using the CIELAB method and above 80% using the flake sorting method. • All de-inking samples exhibited a reduced melt flow index, indicating degradation during pretreatment. • Bio-based, BIO 691 surfactant by Lankem Ltd is used for the study at pH 8. • A factorial design analysis is conducted to model the experiment. • Bath temperature is identified as the most critical factor in de-inking. • De-inking performance also improved with surfactant concentration. • Bath time and brushing iterations had no significant impact on de-inking. • Complete de-inking achieved at 40°C and 5% surfactant concentration. • As solvents (organic solvents or water with alcohols) evaporate during the printing process, it has a very low influence on the de-inking process. • Multiple combinations of pigments and binders used in common plastic printing are evaluated. • In comparing the different pigment types, the highest deinking was observed at 1 wt% NaOH, 55 °C and 0.25 wt% surfactant for all pigments, except black. This is possibly due to the highly hydrophobic carbon black possessing a low HLB value. • For inks with NC binders, de-inking happened through solubilisation, whereas for PVB and UV binders, de-inking happened through peeling. • OPV limits the de-inking efficiency due to the cross-linked structure.
[35]	PE film	Solvent-based	FTIR, Zeta-potential, AFM, SEM, H NMR, FE-SEM	Foam properties, zeta-potential, kraft point, emulsifying power, CMC, temperature, pH	
[12]	Post-consumer LDPE films	Solvent-based	Melt flow index, DSC, tensile tests	pH,	
[69]*	PE films	Solvent-based	-	Concentration, temperature, time, brushing	
[13]	LDPE films	-	Photocolorimeter	Pigment types, binder types, OPV types	

(continued on next page)

Table D.1 (continued)

Source	Plastic type	Ink type	Tests conducted	Parameters evaluated	Observations
[39]	Post-industrial Multilayer plastic film	PU based ink	ATR-FTIR, HT-GPC, DSC, TGA, Spectrophotometer		• Solvent-Targeted Recovery and Precipitation (STRAP) was employed. • Dodecane dissolved PE fraction. • DMSO dissolved EVOH fraction. • Gamma-valerolactone (GVL) removes ink from PET. However sequential steps, longer de-inking time and/or high temperatures were required to improve de-inking performance. • After STRAP, certain decreases in crystallinity were observed for EVOH and PET, possibly due to contamination. • The STRAP process becomes economically viable above 3000 tonnes per year. • Virgin film production was found to have 51% more environmental emissions with regards to the STRAP process, inclusive of the de-inking step. • Solubility and liquefaction of 14 different polymer resins were identified in 7 different liquid media. • PU resins exhibited the lowest solubility due to high chemical resistance.
[11]	LDPE, PP	Water-based	UV/VIS spectroscopy, TGA-FTIR, Reflection Densitometry	Resin solubility, resin liquefaction,	• Acid-based media is more effective in liquefying a broader range of polymer resins. • Highest de-inking efficiency (>60%) was attained in acid-based media with 5 min of interaction. • Acetone showed good de-inking efficiency (~80%) for NC and PU resins within 10 min of interaction.

AFM - Atomic force microscopy; CNF - Cellulose nanofibers; cPAM - Cationic polyacrylamide; CTAB - Hexadecyl trimethylammonium bromide; DMSO - Dimethyl sulfoxide; DSC - Differential scanning calorimetry; DTAB - Dodecyl trimethylammonium bromide; EDX - Energy dispersive X-ray analysis; EVOH - Ethylene vinyl alcohol; FTIR-Fourier transform infrared spectroscopy; GD-Pyr-Br & GT-Pyr-Br - Recycled PET based cationic surfactants; H NMR - Proton nuclear magnetic resonance; HT-GPC - High temperature gel permeation chromatography; SEM - Scanning electron microscopy; TEM - Transmission electron microscopy; TGA - Thermogravimetric analysis; TTAB - Tetradecyl trimethylammonium bromide; WCOEQ - Cationic surfactant from waste cooking oil

*Two studies that did not meet the inclusion criteria for the literature review yet focused on de-inking were added to the table for a comprehensive literature overview.

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